Intermediate Elective 2

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```r
# general use
library(here)

library(tidyverse)

library(janitor)

library(snakecase)

# packages for visualization customization

library(scales)
library(paletteer)
library(ggthemes)
library(RColorBrewer)

# reading in .xlsx files
library(readxl)
```

```r
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")
```

```r
# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
 ↪  Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

#1. Explain

- I chose the most popular Ikea designers visualization because I felt that it would be a good type of visualization to showcase the most abundance invertebrate species at NCOS.
- This visualization makes sense for this the invertebrate visualization because there are 15+ different designers, which is similar to how many different taxa there are. And the different sized bars can account for differences in abundance.
- All Taxa will be shown in my visualization. There will be a different color for each taxon, and the amount of the color will show how abundant they show up in samples.

#2. Plan

```r
# adds in jpeg
knitr::include_graphics(here("data", "plan.jpeg"))
```

#3. Code

```r
# creating new object from sites
ncos_sites <- sites |>

  # filter to only include NCOS quarterly sampling sites
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```r
water_quality_clean <- water_quality |>

  # cleaning column names
  clean_names() |>

  # filtering join: only include sites in ncos_sites
  # keep all the rows on the left that match up with rows on the right
  semi_join(ncos_sites, by = "site") |>

  # combine date and site together and remove those indiv. columns
  unite("date_site", date, site, remove = TRUE) |>

  # group by date_site
  group_by(date_site) |>

  # create new columns that show median of each variable at each date_site
  # because there are multiple samples at the same site at different depths
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>

  # ungroup
  ungroup() |>

  # filter out the salinity outlier
  filter(date_site != "2022-08-12_NVBR")
```

```r
# creating new object from taxon_list
taxon_list_clean <- taxon_list |>

  # converted taxon name values to snake case (lowercase)
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))
```

```
# creating new clean aquatic inverts object from aquatic inverts
aq_ins_clean <- aq_ins |>

  # cleaning column names
  clean_names() |>

  # filtering join: only include quarterly NCOS surveys from ncos_sites
  semi_join(ncos_sites, by = "site") |>

  # renaming column to just date
  rename(date = date_on_vial) |>

  # select columns of interest
  select(date, sample_type, site,

         # 9 most abundant taxon
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>

  # filter to only include filtered beaker samples
  filter(sample_type %in% c("FB250", "FB 250"))  |>


  pivot_longer(
    # cols = columns to make longer
    cols = ostracod:annelida_polychaete,
    # new column for the old column names
    names_to = "taxon",
    # new column for the values in each of the old columns
    values_to = "abundance"
  ) |>

  # group by the date, site, and taxon
  group_by(date, site, taxon) |>

  # calculate average abundance per liter (sum abundance divided by 7.5
   ↪  liters)
```

```r
  summarize(avg_per_lit = sum(abundance, na.rm = TRUE)/7.5) |>

  # ungroup
  ungroup() |>

   # create a new column called `date_site` to join with water quality
   # don't remove the original columns
  unite("date_site", date, site, remove = FALSE) |>

  # join with water quality clean
  # wqc has `date_site` column as well
  left_join(water_quality_clean, by = "date_site") |>

  # join with taxonomic information from taxon list
  # verbatim_name column in taxon_list_clean matches with taxon column in
   ↪  aq_ins_clean
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>

  # fill in full site names
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "East Channel",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  ))  |>

   # setting factor levels for full site name by median salinity
   mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
    ↪  = TRUE),
          site_full = fct_rev(site_full))
```

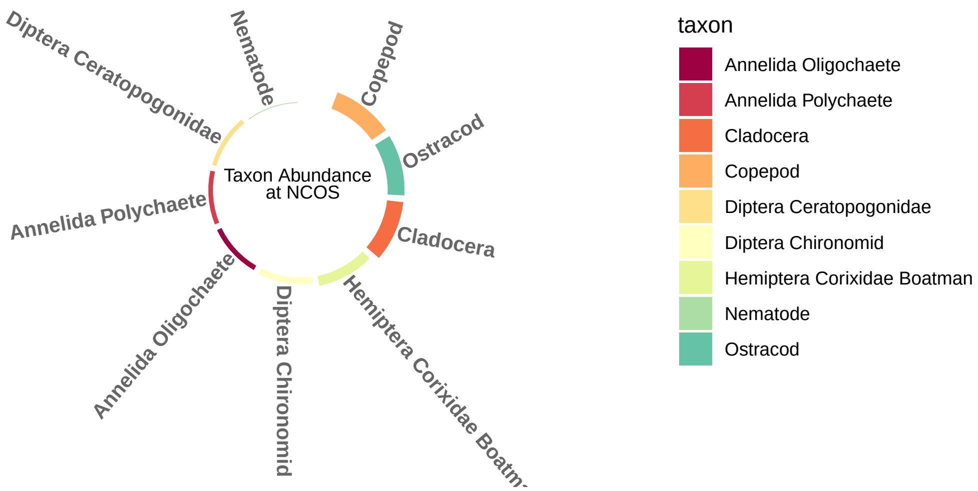
```r
# create new object
data <- aq_ins_clean |>

  # select only taxon and average
  select(taxon, avg_per_lit) |>

  # group by taxon
```

```r
  group_by(taxon) |>

  # sum up all averages
  mutate(total_avg_abundance = sum(avg_per_lit)) |>

  # ungroup
  ungroup() |>

  # transform the total average per liter
  mutate(log_total_avg = log(total_avg_abundance + 1)) |>

  # keep taxon and transformed log
  select(taxon, log_total_avg) |>

  # keep unique rows
  unique() |>

  # change taxon values to title case
   mutate(taxon = str_replace_all(taxon, "_", " ") |> str_to_title())

# set factors
data <- mutate_at(data, vars(taxon), as.factor)
```

```r
# reorder data df and set a new row id
data <- data[order(-data$log_total_avg), ]
data$id <- seq(1, nrow(data))

# copied from inspiration R file
# set up visualization
# Get the name and the y position of each label
label_data <- data
number_of_bar <- nrow(label_data)
angle <- 90 - 360 * (label_data$id-0.5) /number_of_bar
label_data$hjust <- ifelse(angle < -90, 1, 0)
label_data$angle <- ifelse(angle < -90, angle+180, angle)

pal <- brewer.pal(11, "Spectral")

# end of copied code
#
# Plotting
p <- ggplot(data)+
```

```r
geom_bar(aes(x=as.factor(id), y=log_total_avg, fill=taxon),
 ↪  stat="identity") +

# colors
scale_fill_manual(values=pal)+

# setting dimensions
ylim(-40,90) +

# setting theme
theme_bw() +
theme_minimal() +


theme(
  axis.text = element_blank(),
  axis.title = element_blank(),
  panel.grid = element_blank(),
  plot.margin = unit(rep(-1,4,), "cm"),
  legend.margin = margin(0, 2, 0, 0)) +

# setting labels
annotate(geom = "text",
         x=0,y= -30,
         hjust=.5, vjust=1,
         label="Taxon Abundance \n at NCOS",
         size=3, lineheight=.8,
         color="black") +

annotate(geom = "text",
         x=0,y=200,
         vjust=1,
         hjust=.5,
         label = "Data:  Kaggle & TidyTuesday | Graphic: Cristina Cametti -
          ↪  @CameCry",
         size=2.5,
         color="black") +

coord_polar(start = 0)  +

# setting labels
geom_text(data=label_data, aes(x=id, y=log_total_avg + 1, label= taxon,
 ↪  hjust=hjust), color="black",
```

```
            fontface="bold",alpha=0.6, size=3.4, angle=label_data$angle,
             ↪  inherit.aes = FALSE )

# display plot
p
```



4. Process

- The process of using someone else's code was a little difficult as I had to go through each line of code to see what the code did.
- A challenge I encountered was the lack of comments in the R file that I used.
- This visualization is different since it shows just one variable in bars in a circular shape without any axes.